Li-Ion Battery Technology

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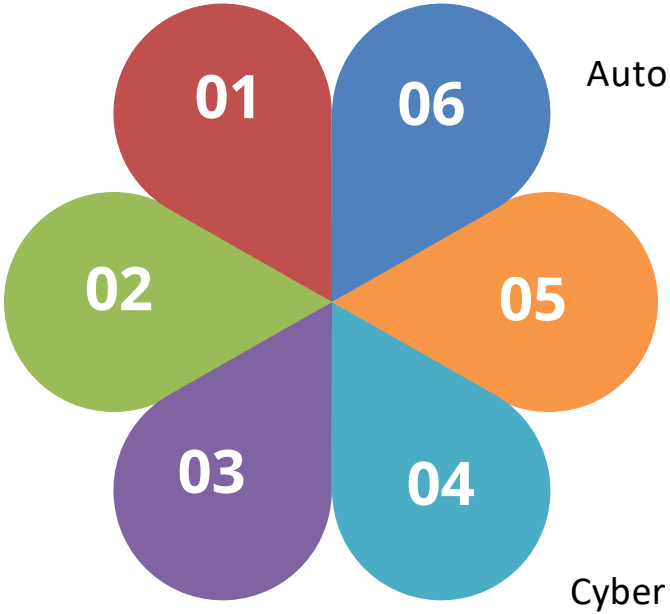
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Our Expertise

Electric Vehicle Technology

Future Mobility Overview

ADAS & Autonomous Technology



Automotive Electronics

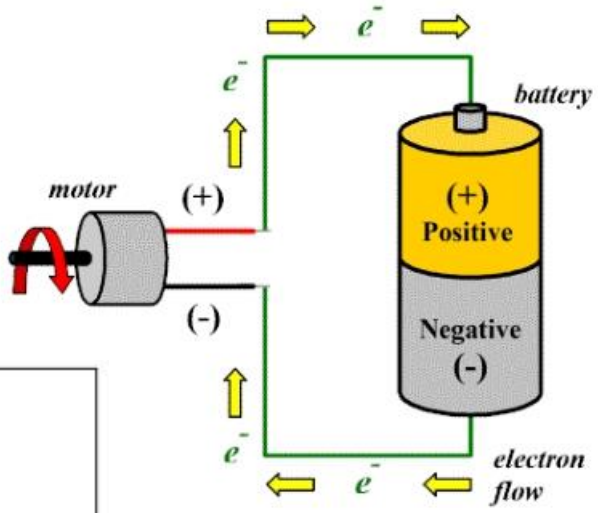
IoT, Connectivity & Advanced Computing

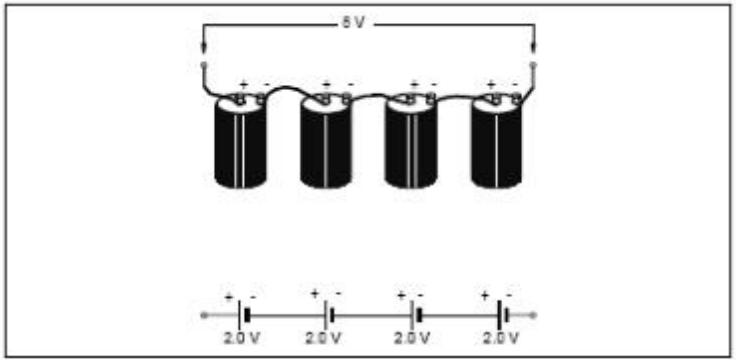
Cyber Security for Automotive

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Battery

- Convert stored chemical energy into electrical energy and vice versa
- Consisting of multiple electrochemical cells
- Contains
  - Electrodes
  - Electrolyte
  - Separator

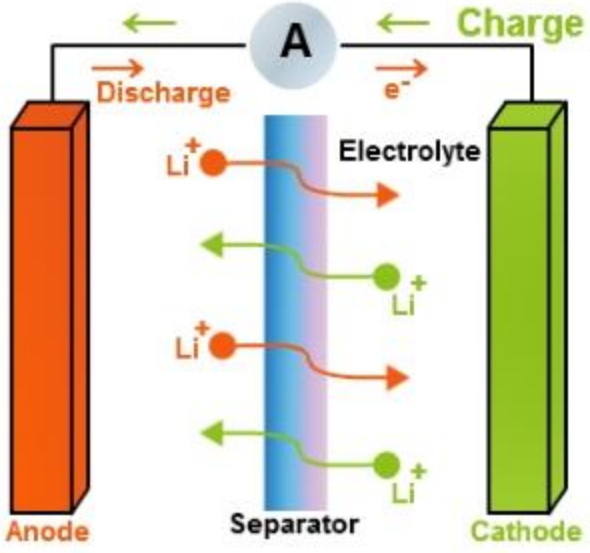




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Electrodes and Electrolytes

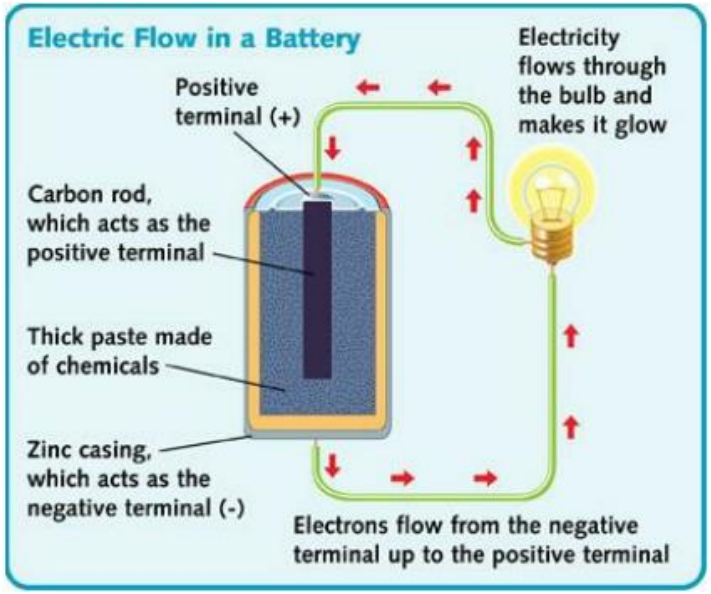
- Cathode
  - Positive terminal
  - Chemical reduction occurs (gain electrons)
- Anode
  - Negative terminal
  - Chemical oxidation occurs (lose electrons)
- Electrolytes allow:
  - Separation of ionic transport and electrical transport
  - Ions to move between electrodes and terminals
  - Current to flow out of the battery to perform work



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Types of Batteries: Primary Batteries

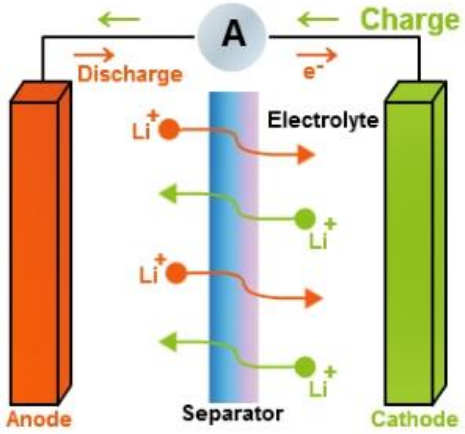
- One use (non-rechargeable)
- Chemical reaction cannot be reversed
- Used when long periods of storage are required
- Lower discharge rate
- Use: - Torch, remote controls etc.



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Types of Batteries: Secondary / Rechargeable

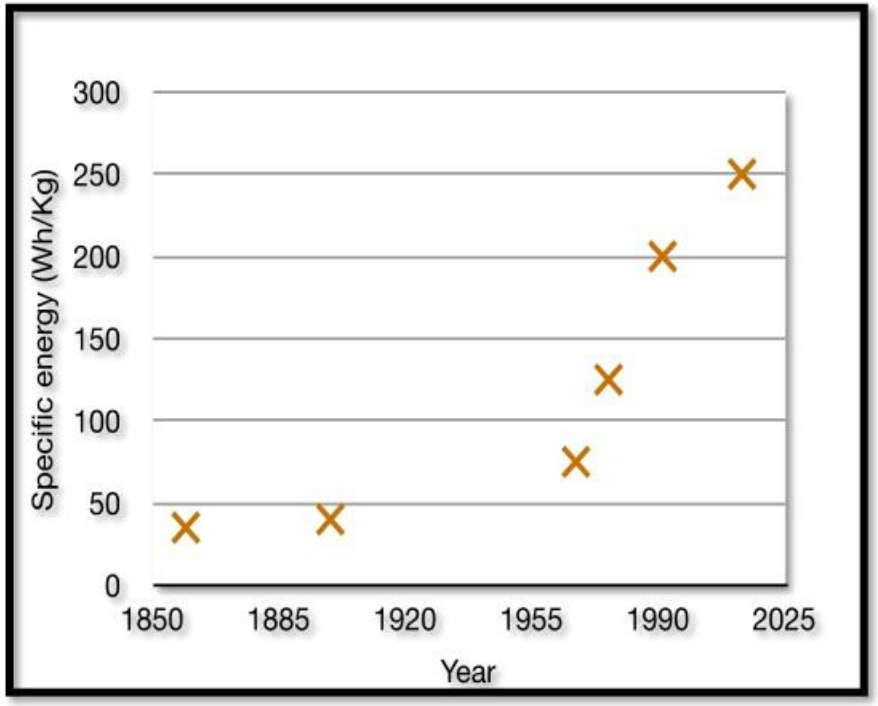
- Reaction can be readily reversed
- Similar to primary cells except redox reaction can be reversed
- Recharging:
  - Electrodes undergo the opposite process than discharging
  - Cathode is oxidized and produces electrons
  - Electrons absorbed by anode
- Types:- Pb-acid, Ni-Cd, Ni-MH, Lithium Li-ion



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Evolution of secondary batteries

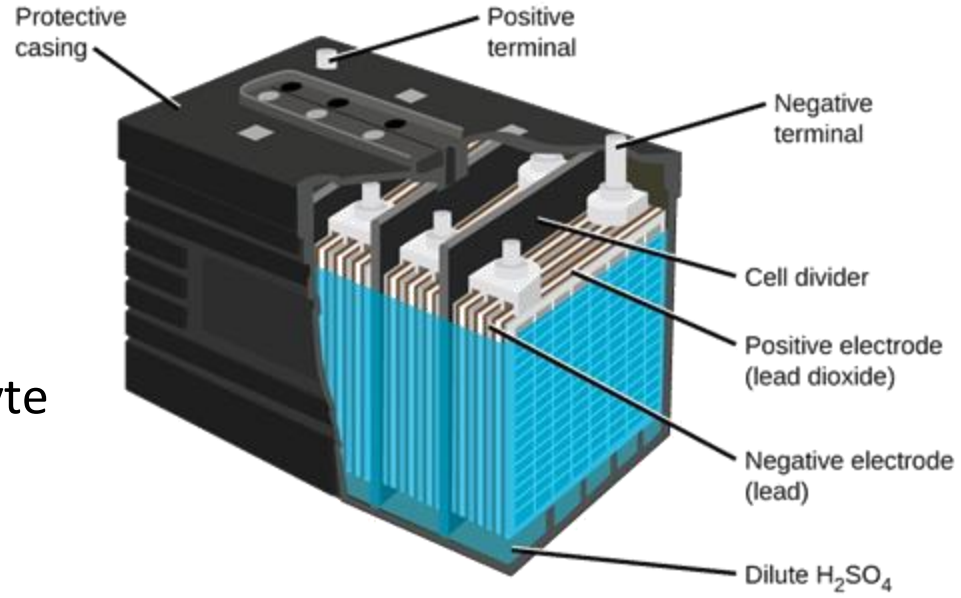
Chemistry	Year	Specific Energy
Lead-Acid	1859	35 Wh/kg
Ni-Cd	1899	40 Wh/kg
Ni-MH	1967	75 Wh/kg
Lithium based	1976	125 Wh/kg
Li-ion	1991	200 Wh/kg
Advanced Li-ion	2013	250 Wh/kg



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Lead Acid Batteries

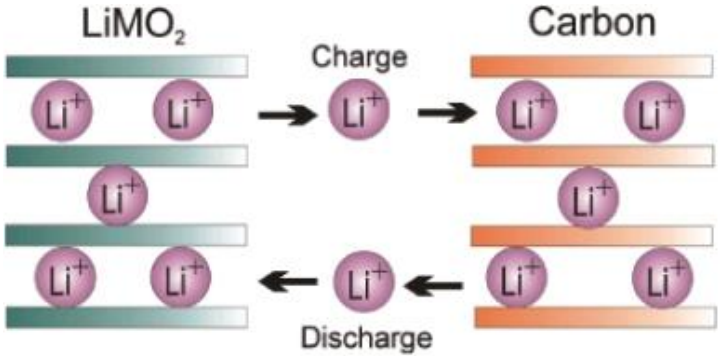
- $Pb\text{-}H_2SO_4\text{-}PbO_2$
  - Gaston Planté, 1859
  - Lead and lead dioxide (PbO_2)
  - Sulfuric acid (H_2SO_4) as electrolyte
  - First reversible battery



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Lithium-ion Batteries

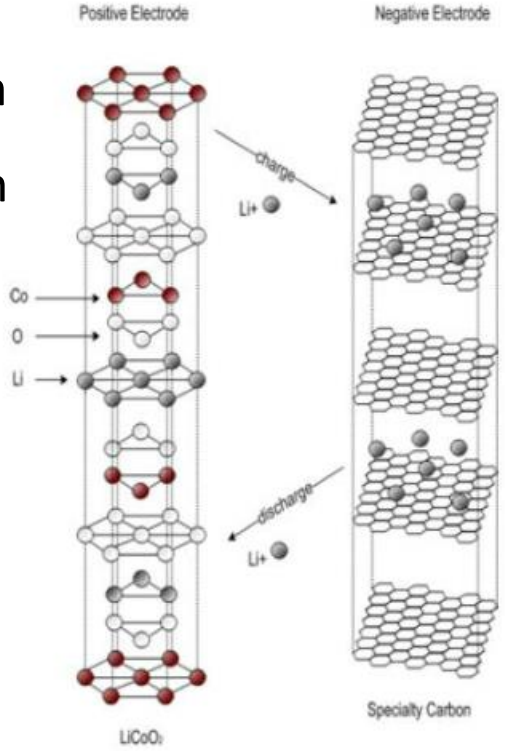
- Secondary battery uses intercalation of Li^+ ions into Carbon electrode
- Commercialized in 1991.
- Lithium ions move from the negative electrode through an electrolyte to the positive electrode during discharge, and back when charging.



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Lithium-ion batteries

- Positive electrode: Lithiated form of a transition metal oxide (lithium cobalt oxide-$LiCoO_2$ or lithium manganese oxide $LiMn_2O_4$)
- Negative electrode: Carbon (C), usually graphite (C_6)
- Electrolyte: solid lithium-salt electrolytes
  1. $LiPF_6$, $LiBF_4$, or $LiClO_4$ and
  2. Organic solvents (ether)



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How do Li-ion battery Work?



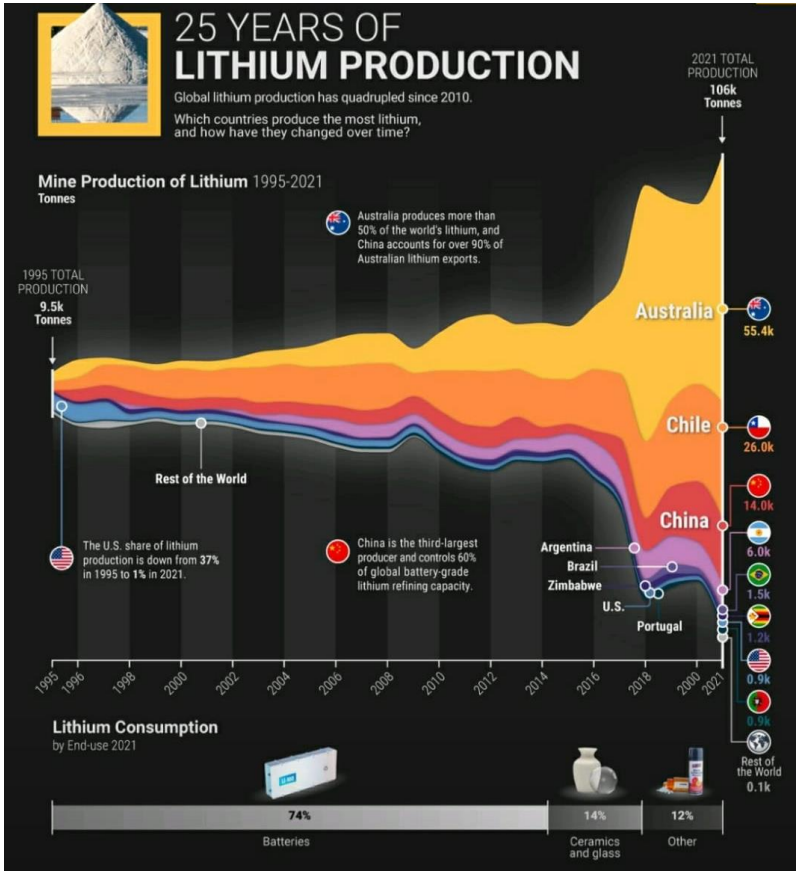
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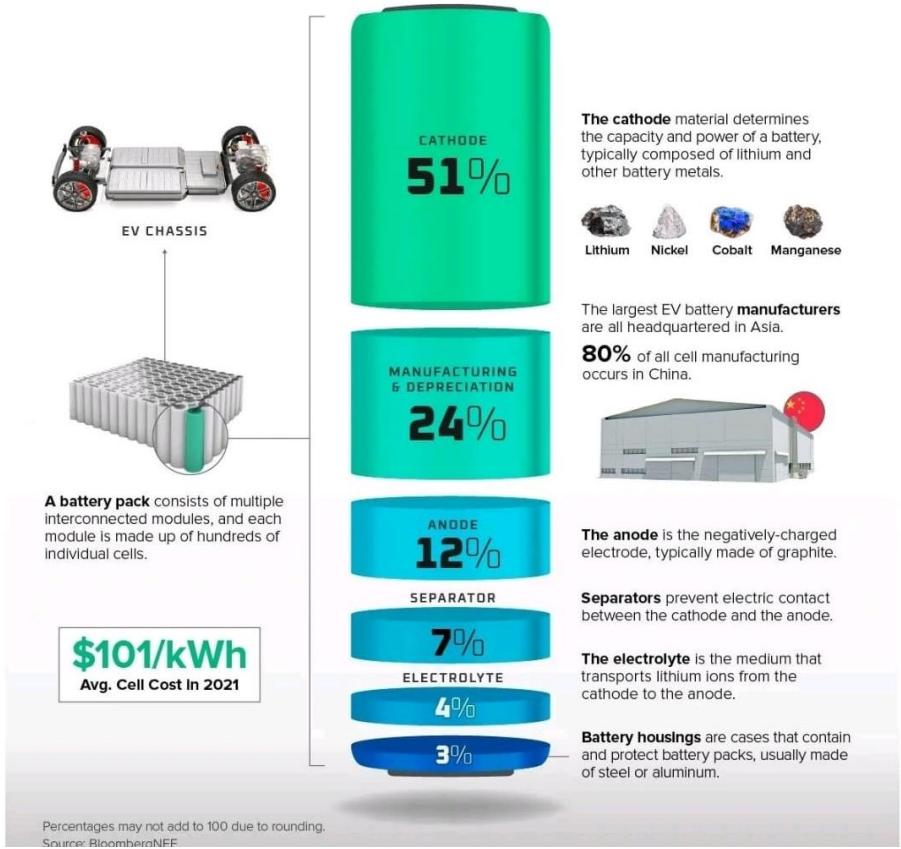
Li-ion Battery - details



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Sources of Lithium and Cost of cell

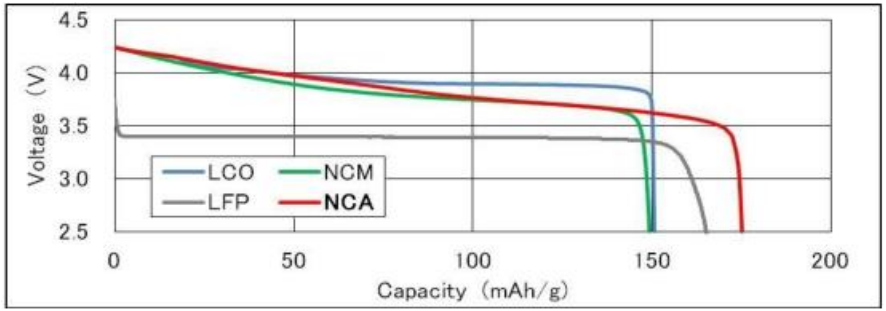




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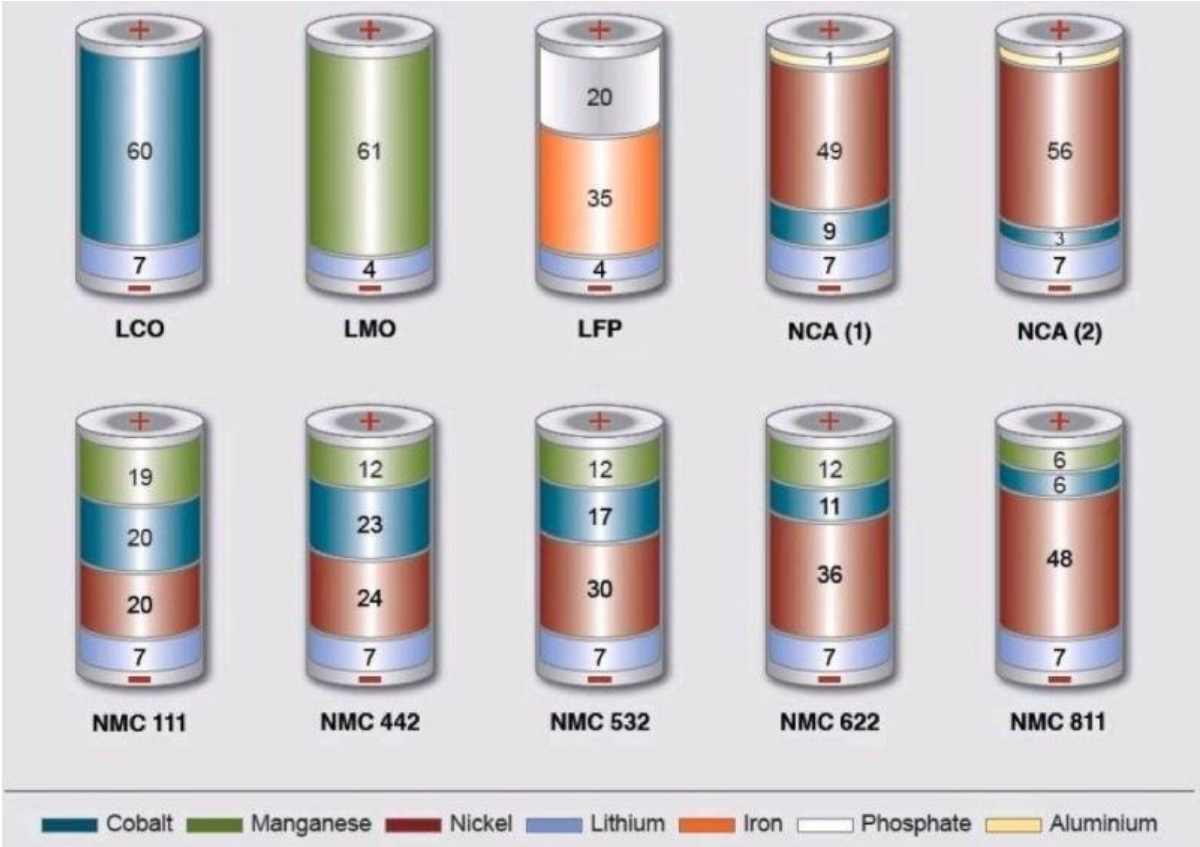
Lithium Ion Chemistries

- **Lithium Nickel Manganese Cobalt Oxide (NCM/NMC cathode)**
- **Lithium Nickel Cobalt Aluminium (NCA cathode)**
- **Lithium Iron Phosphate (LFP cathode)**
- Lithium Cobalt Oxide (LCO cathode)
- Lithium Manganese Oxide spinal (LMO cathode)
- Lithium Sulphur (Li-S, Sulphur cathode & Li Metal anode)
- Lithium Metal (Li Metal anode)
- Lithium Titanate (LTO anode)



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Cathode in different cell chemistries



- The Figure shows metal content in weight % only
- No oxygen content is displayed in the image
- Oxygen content varies from 33-40%
- Only Cathode active materials are displayed

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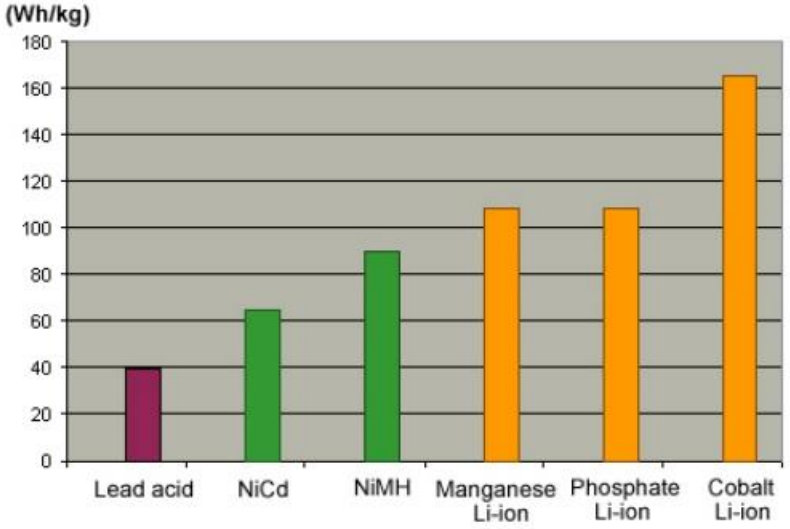
Cell Chemistry - Detailed

Cell Chemistry	Lithium Cobalt Oxide	Lithium Manganese Oxide	Lithium Nickel Manganese Oxide	Lithium Iron Phosphate	Lithium Nickel Cobalt Aluminum Oxide	Lithium Titanate Oxide
Short form	Li-cobalt	Li-manganese	NMC	Li-phosphate	Li-aluminum	Li-titanate
Abbreviation	$LiCoO_2$ (LCO)	$LiMn_2O_4$ (LMO)	$LiNiMnCoO_2$ (NMC)	$LiFePO_4$ (LFP)	$LiNiCoAlO_2$ (NCA)	Li_2TiO_3 (common) (LTO)
Nominal voltage	3.60V	3.70V (3.80V)	3.60V (3.70V)	3.20, 3.30V	3.60V	2.40V
Full charge	4.20V	4.20V	4.20V (or higher)	3.65V	4.20V	2.85V
Full discharge	3.00V	3.00V	3.00V	2.50V	3.00V	1.80V
Minimal voltage	2.50V	2.50V	2.50V	2.00V	2.50V	1.50V (est.)
Specific Energy	150–200Wh/kg	100–150Wh/kg	150–220Wh/kg	90–120Wh/kg	200-260Wh/kg	70–80Wh/kg
Charge rate	0.7–1C (3h)	0.7–1C (3h)	0.7–1C (3h)	1C (3h)	1C	1C (5C max)
Discharge rate	1C (1h)	1C, 10C possible	1–2C	1C (25C pule)	1C	10C possible
Cycle life (ideal)	500–1000	300–700	1000–2000	1000–2000	500	3,000–7,000
Thermal runaway	150°C (higher when empty)	250°C (higher when empty)	210°C(higher when empty)	270°C (safe at full charge)	150°C (higher when empty)	One of safest Li-ion batteries

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Specific Energy (Gravimetric Energy Density)

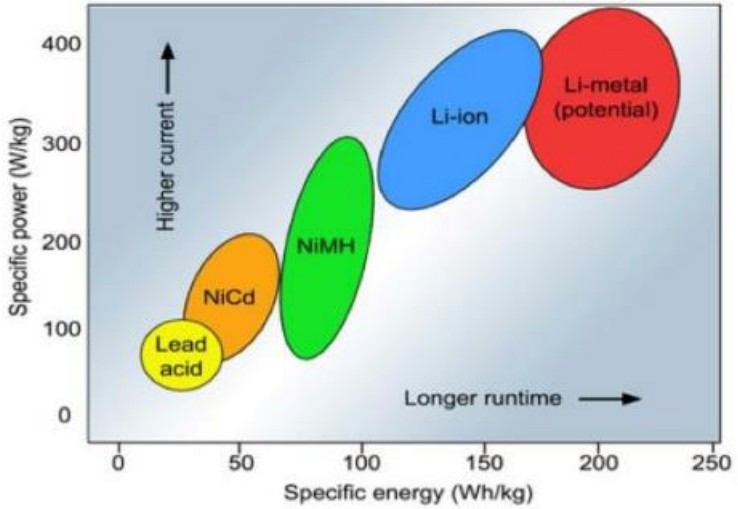
- The amount of electric energy a battery can hold per unit mass
- It normally has units of Wh/Kg



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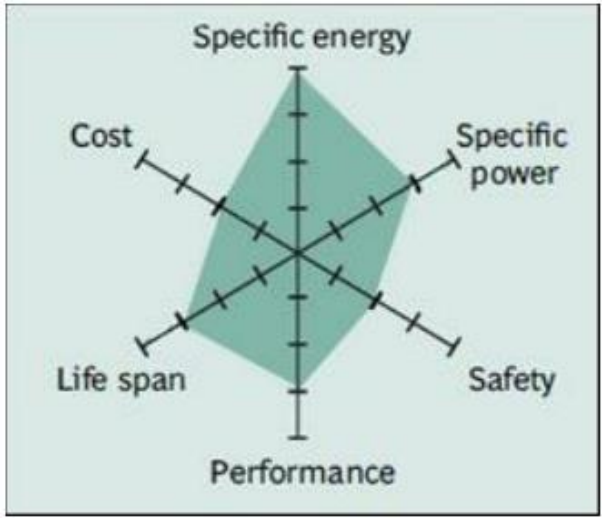
Power Density

- Power density is the ability of the battery to take on/deliver power per unit mass of the battery.
- It normally has units of W/Kg .

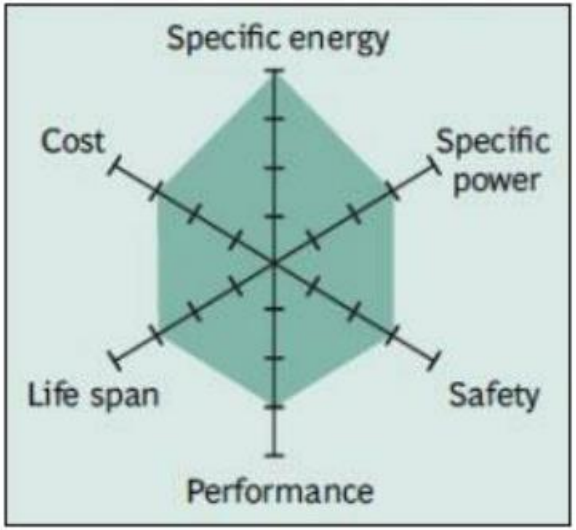


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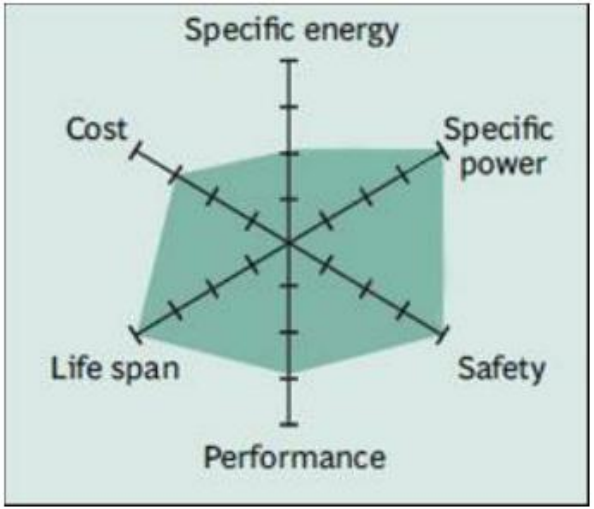
Comparing Chemistries



NCA – Li-Ni-Co-Al Cathode chemistry



NMC – Li-Ni-Mn-Co Cathode chemistry

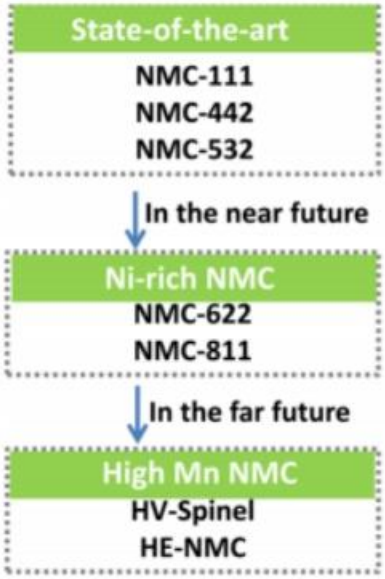


LFP – Li-Fe-PO_4 Cathode chemistry

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Types of NMC

- **NMC:** The cathode combination is typically one- third nickel, one-third manganese and one-third cobalt, also known as 1-1-1.
- **NCM:** 5 parts nickel, 3 parts cobalt and 2 parts manganese (5-3-2)
- There are also other types like NMC-422, NMC- 622, NMC-811 etc., depending on different concentrations of Ni, Co & Mn.
- The aim of research in this field is to reduce dependency on expensive and difficult to procure materials.



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Selection of Cell chemistry

- Selection of a particular chemistry depends on
  1. Specific energy density (Wh/kg)
  2. Volumetric energy density (Wh/l)
  3. Life cycle at certain C-rate,
  4. Temperature at certain depth of discharge (DoD)
  5. Capacity and DoD
  6. C-rate usable
  7. Safety and safe disposal
  8. Cost/kWh

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Ideal Batteries

- Higher energy per kg [≥ 250 Wh/kg]
- Long cycle life [1500-3000 cycles]
- Deep discharge [≥ 95%]
- Highly efficient [≥ 99%]
- Fast charge [≥ 2C]
- Flat discharge curve
- Low self discharge
- Maintenance free
- Not toxic
- No disposal problems

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Voltage & Current ; Capacity & Power

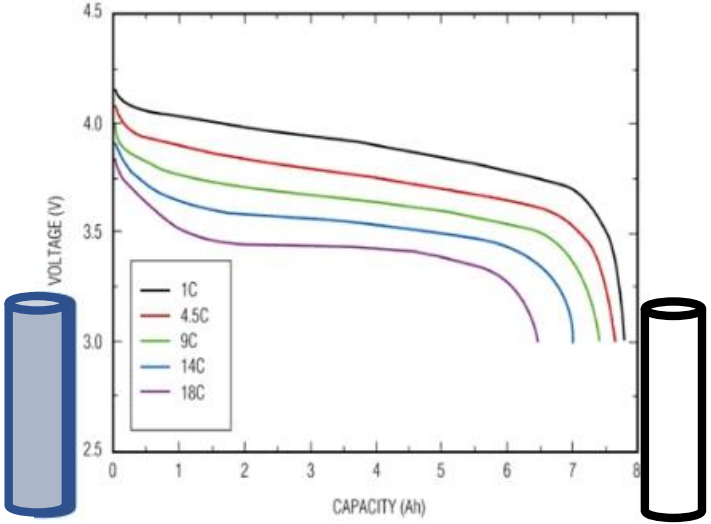
- Nominal voltage; Cell over voltage; Cell under voltage; Cell end of life voltage; Nominal current

Capacity (Ah) = Current (A) × Time (H)

- Total charge (no. of electrons) which can be delivered by a cell or battery under specified conditions.

Power (W) = Current (A) × Voltage (V)

- The rate of energy supplied.



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Battery Capacity

The amount of energy available from a battery.

- **Chemical Capacity** – full storage capacity of the chemistry when measured from full to empty or empty to full. This is normally defined at a given C-rate and maximum and minimum voltages.
- **Designed Capacity** – the storage capacity allowed to be used by the application. Often called the Usable SoC Window for the battery pack. This is designed to ensure safety, performance and longevity.
- **Available Capacity** – this is the capacity that can be accessed taking into account the temperature, age, health, and use of the cell.

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Rated Capacity

Battery capacity is expressed in ampere-hours (Ah) and is effected by:

- Temperature
- Discharge rate – normally the higher the discharge rate the lower the capacity.
- Ageing – capacity will decrease with calendar life and based on the usage history.

- Under well defined conditions this is often referred to as the **Rated Capacity.**
- Battery capacity is likely to be different under different temperature, discharge rates and prior use.

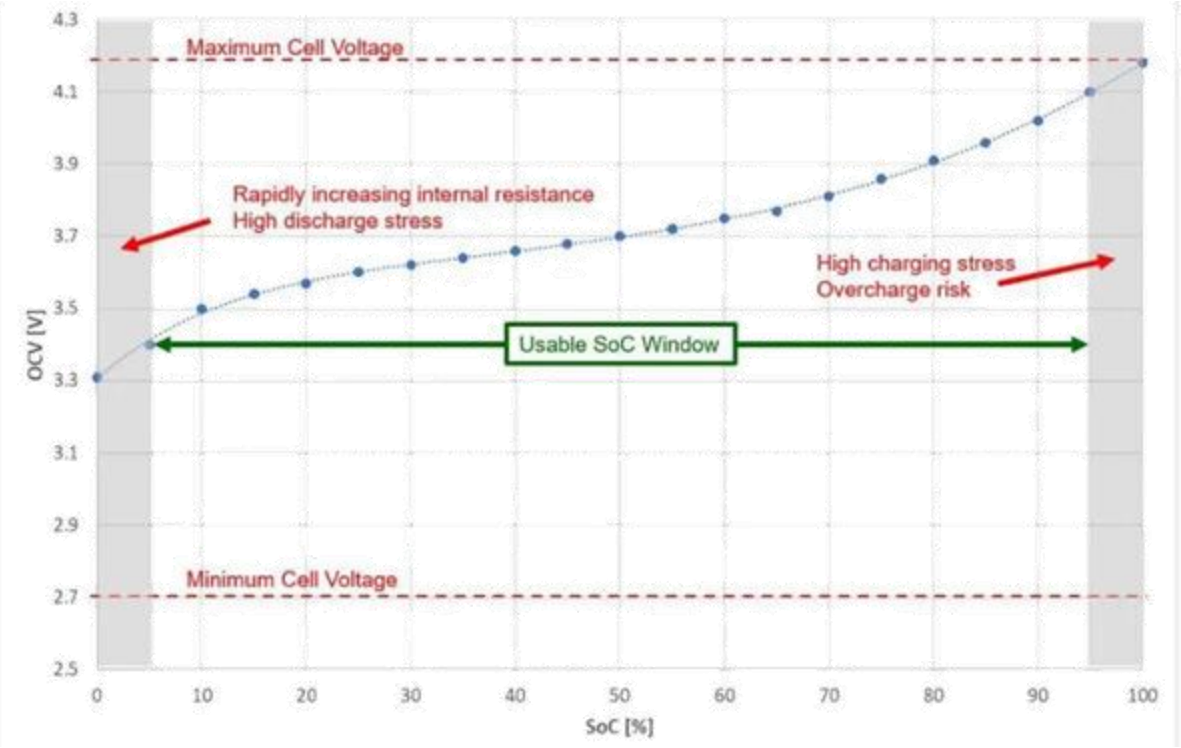
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Usable SoC window

- For the battery cell to last a lot of cycles, extend the life or to ensure the available power is consistent; we need to set a usable SoC window that is smaller than 100%.
- We might limit the top end charge to perhaps 95% SoC and the bottom end discharge to 5% SoC.
- High SoC and hence high cell voltage stresses the cell and significantly reduces the lifetime.
- At low SoC, the Open Circuit Voltage (OCV) is decreasing and the internal resistance of the cell increasing.

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Usable SoC window



- The downside is the usable energy has just been reduced by 10%

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Usable capacity

- For higher power packs the usable SoC window will be even smaller. A high power hybrid battery pack may have a usable SoC window of just 30%.

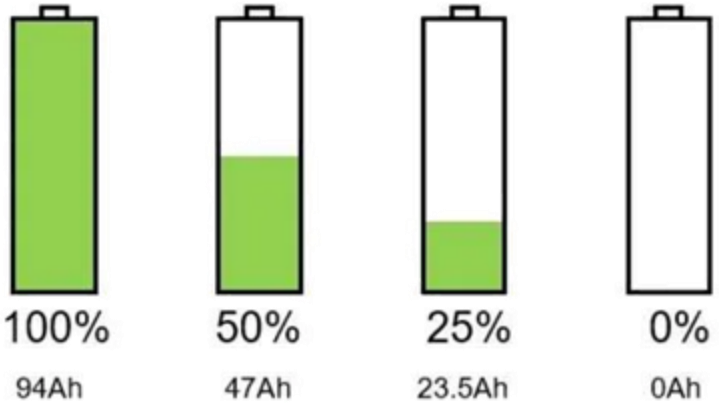
Vehicle	Total Capacity	Usable Capacity	%
Mercedes EQA 250	69.7kWh	66.5kWh	95.4%
Nissan Leaf	40kWh	37kWh	92.5%
Ford F-150 Lightning	142kWh	131kWh	92%
Porsche Taycan	93.4kWh	83.7kWh	90%
Renault Zoe ZE 40	45.3kWh	41kWh	90%
BMW 330e PHEV	12kWh	9.8kWh	82%
Ford C-max Hybrid	1.4kWh	0.56kWh	40%

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State of Charge

- State of Charge (SoC) and defined as the amount of charge in the cell as a percentage compared to the nominal capacity of the cell in Ah.
- As an example, for the Samsung SDI 94Ah cell

$$SoC = \frac{(Ah\ Capacity\ - Current\ X\ time)\ 100}{Ah\ Capacity}$$



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Defining SoC

- The capacity remaining needs to be defined against specific test conditions, e.g. :- rated capacity at C/3 and 25°C.
- The maximum capacity will be established at a maximum cell voltage, e.g. 4.13V for this particular 94Ah cell. (Max cell voltage is 4.5V)
- The 0% is again set under specific conditions and defined by a minimum cell voltage, e.g. 3.3V at C/3 and 25°C. (Minimum cell voltage is 2.7V)
- The SoC estimation of the battery cell is very important as so many other functions depend on the accuracy of this value.

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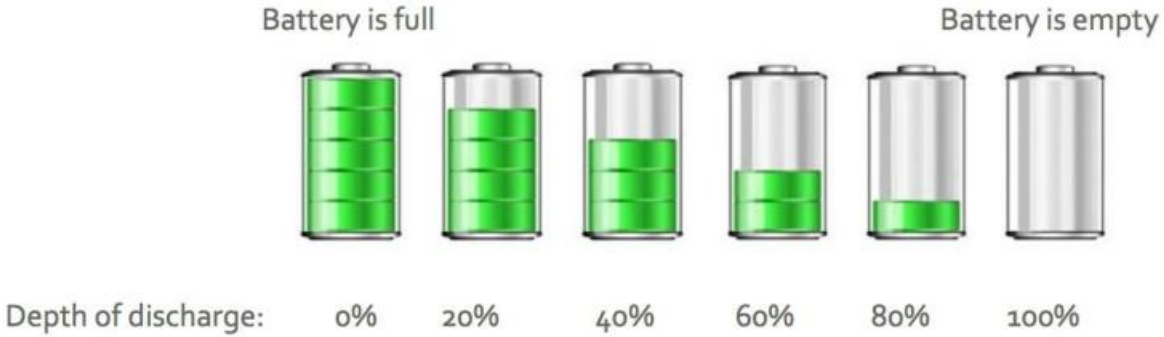
SoC – used for

- SoC is used to estimate a number of parameters, including: maximum charge and discharge current at any instant, the amount of energy left in the battery pack and State of Health.
- SoC is always an estimate as it cannot be directly measured.
- In an operating system the temperature and current can be changing over a large range and sometimes quite rapidly.

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Depth of Discharge

- The ratio of the quantity of electricity removed from the cell or battery on discharge to its capacity.

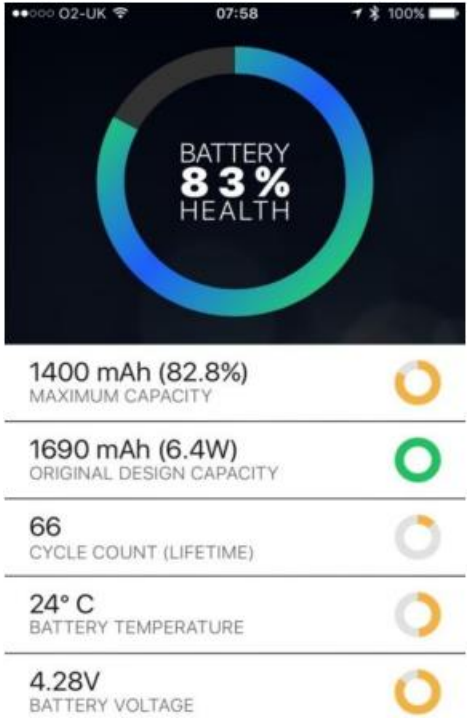


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State of Health

$$SoH = \frac{Present\ Max.Capacity\ (Ah)}{Original\ Capacity\ (Ah)}\ \text{X}\ 100$$

$$= \frac{1400\ mAh}{1690\ mAh}\ X\ 100 = 82.8\%$$



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C - Rate

- C-rate is a measure of the rate at which a battery is discharged relative to its maximum capacity.
- Usually expressed with letter “C”
    - 1C rate means that the discharge current will drain the entire battery in 1 hour.
    - For a battery with a capacity of 100 Amp-hrs, this equates to a discharge current of 100 Amps.
    - 5C rate for this battery would be 500 Amps, and a C/2 rate would be 50 Amps.

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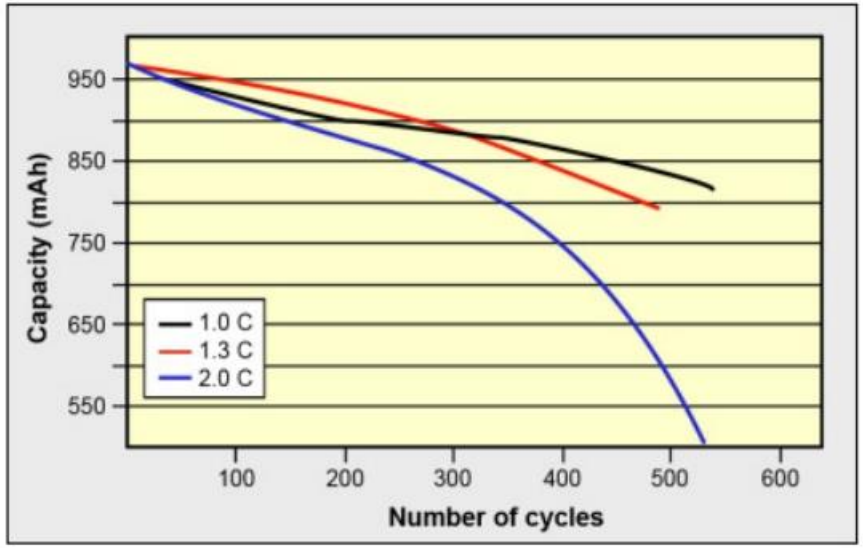
Battery Efficiency

- **Columbic Efficiency**
  - The columbic efficiency of battery the ratio of the number of charges that enter the battery during charging compared to the number that can be extracted from the battery during discharging.
  - In general, the columbic efficiency may be high, in excess of 95%.

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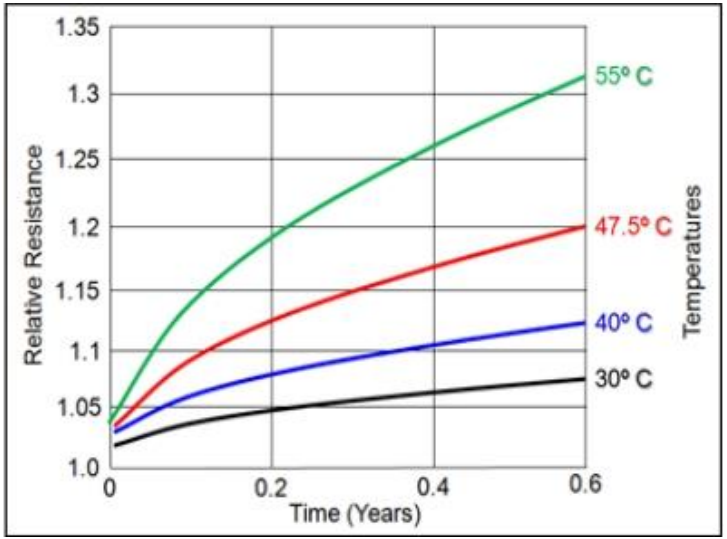
Lifetime

Cycle Life



The number of complete charge-discharge cycles.

Shelf Life



The time an inactive battery can be stored for.

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Vehicle Cycle Life

QUESTION:

- Find out which vehicle battery has to be replaced 1st and life of battery in days or years. Both vehicles travel 50km per day.

Vehicle	Range	Cell Rates Cycle Life
Vehicle A	600 KM	500 Cycles
Vehicle B	100 KM	2000 Cycles

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Vehicle Cycle Life

QUESTION:

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Vehicle	Range	Cell Rates Cycle Life
Vehicle A	600 KM	500 Cycles
Vehicle B	100 KM	2000 Cycles

- A – 600/50 = 12 days for recharge
  - 12X500 = 6000 days = **16.4 years**
- B – 100/2 = 2 days for recharge
  - 2X2000 = 4000 days = **10.9 years**

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